Rachit Kale

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Education

K. J. Somaiya School of Engineering (KJSSE), Mumbai *2023 – 2027*

CGPA: 7.52

Final Year Project: Hybrid Anomaly Detection for Identifying Ambiguous UPI Transactions

Experience

Software Developer Intern — Synoris Information Systems Pvt Ltd. *Jun 2026 – Aug 2026*

Surat, India

MERN, Tailwind CSS, WebSockets, JWT auth, Postman, Gemini GenAI, GitHub

- Proposed and engineered a **full-stack UPI transaction tracking system** that intercepted and logged UPI payments in **real time**, eliminating manual transaction entry entirely; approach was selected over alternative team proposals for its **lower implementation complexity and faster time to ship**
- Integrated **Gemini GenAI** into the transaction pipeline to auto classify payments across **9 predefined spending categories**, powering a downstream analytics layer for automated report generation, category wise trend analysis and predictive spend suggestions
- Validated API integrations across frontend, backend, and AI service layers using **Postman**; collaborated with the project lead through iterative reviews, using Git for **feature branch based development**

Software Development Intern — KJSSE (Institutional R&D) *Jan 2026 – Apr 2026*

Mumbai, India

MERN, Gemini API, GitHub, Vercel, Web Speech API, Monaco Editor, JWT auth

- Developed a **full-stack multimodal interview prep platform using MERN**, integrating the **Gemini API** to dynamically generate technical and behavioral questions based on user role, skills and uploaded resume
- Designed and contributed to RESTful APIs architecture for session management and automated feedback generation, **reducing feedback response time from 15s to 3-5s** through response optimization; collaborated on system architecture and testing for scalable deployment
- Built voice, video and coding based interview simulation modules with real time interaction, speech-to-text processing and automated feedback with **readiness score evaluation**

Projects

Hybrid Anomaly Detection for Ambiguous UPI Transactions *Feb 2026 – Present*

React, Python, Apache Parquet, XGBoost, Random Forest, Isolation Forest, FastAPI

- Architectured a hybrid fraud detection pipeline combining **supervised models (XGBoost, Random Forest)** with **unsupervised anomaly detection (Isolation Forest, LOF)** across 4 real world datasets, engineering a unified schema from heterogeneous transaction sources
- Designed an **explainable fusion scoring layer** combining dual model signals and uncertainty metrics to **flag ambiguous transactions** for review, avoiding a third opaque classifier
- Built a **React & FastAPI** testing dashboard for transaction level explainability, distribution based finance reports and browser persisted prediction logs for manual model inspection

AI-Driven Train Induction Planning & Scheduling System *Sep 2025 – Nov 2025*

React.js, Node.js, PostgreSQL, AI Simulation, Data Visualization, DeepSeek API

- Designed and developed a interactive dashboard based decision support system for automated train induction planning for KMRL
- Developed an AI based simulation module to generate optimised train induction schedules using rule based constraints and multi objective logic
- Implemented explainable output generation with conflict alerts and "what-if" simulation capabilities
- Collaborated in a cross functional team environment to translate complex operational requirements into a scalable digital solution

Technical Skills

Languages: JavaScript, Typescript, SQL, HTML5, CSS3
Backend & Database: Node.js, PostgreSQL, MongoDB
Frontend: React.js, HTML, CSS, Tailwind CSS
Tools: Git, GitHub

Certifications & Achievements

- Qualified for Round 2 — Smart India Hackathon 2025 (National Level); Developed AI-Driven Train Induction Planning & Scheduling system for Kochi Metro Rail Limited